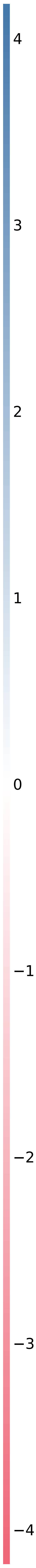
Question:
A 34-year-old woman presents to her OB/GYN with complaints of missing her last 3 periods as well as intermittent spontaneous milky-white nipple discharge bilaterally for the past 3 months. Vital signs are stable and within normal limits. Neurologic examination is without abnormality, including normal visual fields. Serology and MRI of the brain are ordered, with results pending. Which of the following sets of laboratory results would be expected in this patient?
Answer Choices:
A. Decreased prolactin, decreased FSH, decreased LH
B. Decreased prolactin, increased FSH, increased LH
C. Increased prolactin, decreased FSH, increased LH
D. Increased prolactin, decreased FSH, decreased LH



Reasoning + Answer:

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Finding Reasoning Paths:
1. **Hormonal imbalance**: Prolactin and gonadotropin levels are central to the endocrine system. Prolactin can be elevated due to various causes, including pituitary adenomas, medications, or systemic conditions. Gonadotropins (FSH and LH) may be elevated or decreased depending on the underlying pathology.
2. **Milky nipple discharge**: This could indicate hyperprolactinemia, which is often associated with pituitary adenomas or medications like cabergoline.
3. **Obstetric and gynecologic presentation**: Amenorrhea and galactorrhea are classic symptoms of hyperprolactinemia, which can be due to pituitary adenomas or medications.
4. **MRI of the brain**: A pituitary adenoma is a common cause of elevated prolactin levels, and imaging may show an adenoma with characteristics like diminished density.
5. **Serologic findings**: Assess whether the patient has other hormone abnormalities (e.g., hypogonadism due to suppression of FSH/LH) or systemic conditions (e.g., hypothyroidism).

Reasoning Process:
1. **Understanding the symptoms**:
- The patient has amenorrhea and galactorrhea, which are hallmark features of hyperprolactinemia. Hyperprolactinemia can be caused by a pituitary adenoma (prolactinoma), medications (e.g., cabergoline, dopamine agonists), or systemic conditions (e.g., hypothyroidism).

2. **Evaluating the MRI**:
- A pituitary adenoma is highly likely in this case, as it is the most common cause of hyperprolactinemia. MRI findings of a pituitary lesion with a diminished density would confirm a prolactin-secreting adenoma.

3. **Analyzing the hormonal profiles**:
- In a patient with a prolactinoma, prolactin levels are typically elevated. Gonadotropins (FSH and LH) are suppressed due to negative feedback inhibition by elevated prolactin. As a result, FSH and LH levels are usually low.
- Thyroid hormones (TSH, T3, T4) and sex hormones (estradiol, testosterone) may be affected depending on the duration and severity of the hyperprolactinemia and associated hypogonadotropic hypogonadism.

4. **Considering other causes**:
- Hypothyroidism can cause elevated prolactin due to decreased feedback inhibition, but the MRI would not typically show a pituitary adenoma in this case.
- Medications like cabergoline or dopamine agonists can cause hyperprolactinemia, but the MRI would not show a pituitary lesion in these cases.

5. **Synthesizing the findings**:
- The patient has amenorrhea, galactorrhea, and likely hyperprolactinemia due to a pituitary adenoma. This would result in:
- **Decreased prolactin levels**: In the absence of feedback inhibition by elevated prolactin.
- **Decreased FSH and LH levels**: Due to negative feedback inhibition by elevated prolactin.
- **Hypothyroidism**: If hypogonadotropic hypogonadism is present, thyroid function tests (e.g., TSH, T3, T4) may show hypothyroidism.

Conclusion:
The laboratory results expected in this patient include decreased prolactin, decreased FSH, and decreased LH. This conclusion aligns with the presence of a pituitary adenoma causing hyperprolactinemia and resulting in decreased FSH and LH due to negative feedback inhibition.
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Increased prolactin, decreased F